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The Editors
Journal of Cosmology and Astroparticle Physics (JCAP)

Dear Editors,

I am writing to submit the manuscript entitled **"The Dark Unification: Self-Interacting Majorana Dark Matter, Path-Integral Foundations, and Dark Energy as the T-Breaking Component"** for consideration in JCAP.

Summary. The paper presents a unified framework for the dark sector built from a single Lagrangian with three free parameters (m_χ, m_ϕ, α). The key results are:

1. **Phenomenology (Part I).** A Majorana fermion scattering via a secluded scalar mediator reproduces velocity-dependent SIDM cross sections that simultaneously satisfy dwarf-galaxy cores, the Bullet Cluster upper limit, rotation-curve diversity, the Radial Acceleration Relation, and relic density — with a global $\chi^2/\nu = 0.38$ over 13 observables and 122 MCMC-identified viable benchmark points.
2. **Path-integral foundations (Part II).** We prove that the Variable Phase Method (VPM) is *exactly* the fluctuation determinant of the one-loop path integral over the mediator field, establishing it as a first-principles result rather than a numerical technique. The Coleman–Weinberg vacuum of the same Lagrangian, combined with dark QCD confinement, predicts $H_0 = 67.4$ km/s/Mpc with no additional free parameters.
3. **Dark electromagnetic duality (Part III).** Promoting the CP phase to a dynamical axion field decomposes the Yukawa coupling into a T-even (dark matter) and T-odd (dark energy) component, unifying DM and DE as projections of a single interaction. The relic angle $\theta_{\text{relic}} = \arctan(1/\sqrt{8}) = 19.47^\circ$ is fixed uniquely by SIDM and relic constraints, predicting $w_0 = -0.727$ — testable by DESI and Euclid within the next data releases.

Why JCAP. The manuscript bridges particle phenomenology, quantum field theory, and observational cosmology — precisely the interdisciplinary scope of JCAP. All predictions are falsifiable, and all numerical code and data are publicly available on Zenodo.

Novelty. To my knowledge, this is the first work that (i) proves VPM = path integral exactly, (ii) derives H_0 from a dark-sector Lagrangian without free parameters, and (iii) unifies dark matter and dark energy as T-even and T-odd projections of the same coupling.

The manuscript has not been submitted elsewhere. I confirm that I have no conflicts of interest.

Sincerely,

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